

<div>Profile</div> <p>M.S. candidate in Chemical Engineering at NTUST, specializing in electrochemistry, ultra-low-temperature electrolytes and lithium battery reliability / aging; co-author on a Nature Energy submission.</p> <div>Research & Industry</div> <div>Long-Life Li Battery Evaluation for Smart Water Meters</div> <div>Partner: a smart-water-meter battery maker</div> <div>Project lead</div> <div>Screening methodology</div> <div>Designed full-stage test protocols, experimental workflows and statistical validation models.</div> <div>Database & SOP</div> <div>Established the test database and standardized high-throughput screening procedures (SOP).</div> <div>Advanced analysis</div> <div>EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.</div> <div>Training & tech transfer</div> <div>Trained QA and R&D teams to independently run large-batch screening and data interpretation.</div> <div>Nature Energy</div> <div>Journal submission — battery thermal stability study</div> <div>Status Under review</div> <div>Co-author</div> <div>Responsible for TGA thermal measurement</div> <div>Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.</div> <div>Experience</div> <div>2025 – Present</div> <div>Sustainable Electrochemical Energy Development (SEED) Center, NTUST</div> <div>Lab Equipment Procurement & Asset Maintenance Manager</div> <div> Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables. Lab safety: monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak. Hazardous materials: handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells. - Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures. TGA testing service: Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's. Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500. </div> <div>Summer 2022</div> <div>New Taipei City Dept. of Urban Development</div> <div>Summer Intern, Housing Development Section</div> <div> - Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes. </div> <div>2021 – 2024</div> <div>Tunghai University Student Housing / GEPT</div> <div>Resident Assistant (2 yrs) / Exam Staff (part-time)</div> <div> - Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors. </div> <div>Conferences & Training</div> <div>Presenter</div> <div>2026 TwICHe Annual Meeting</div> <div>2026/12/05–06 • Feng Chia University</div> <div>Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability</i></div> <div>Vendor seminar</div> <div>TA Instruments ARES-G3 launch & applications forum</div> <div>2026/07/17 • National Taiwan University</div> <div>Vendor lecture</div> <div>TA Instruments TGA lecture</div> <div>2026/06/03 • National Taiwan University</div> <div data-cs="2" data-kind="parent"><div>Skills & Instruments</div><div>Electrochemistry & batteries</div><div>Impedance spectroscopy (EIS)</div><div>Battery aging analysis</div><div>Reliability analysis</div><div>Low-temp electrolytes</div><div>Characterization</div><div>Thermal analysis (TGA / DSC)</div><div>Synchrotron (TLS 20A / TPS 27A / TPS 45A)</div><div>Rheology (ARES-G3)</div><div>Viscometer & densimeter (DMA 5002)</div><div>Process & operations</div><div>Lithography (nanoimprint / self-assembly)</div><div>Equipment procurement & maintenance</div><div>Software</div><div>Aspen Plus V14</div><div>Aspen Dynamics</div><div>Microsoft Office</div><div>OriginPro</div><div>TA Universal Analysis</div><div>TA TRIOS</div><div>Education</div><div>2025 – 2027 (expected)</div><div>National Taiwan University of Science and Technology</div><div>M.S. in Chemical Engineering</div><div>Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.</div><div>2020 – 2024</div><div>Tunghai University</div><div>B.S. in Chemical & Materials Engineering</div><div>Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).</div><div>EHS Licenses</div><div><div><div>✓</div><div>Dedicated Wastewater and Sewage Treatment Specialist (Class B)</div><div>Certified 2026</div></div><div><div>✓</div><div>Indoor Air Quality Maintenance Management Specialist</div><div>Certified 2026</div></div><div><div>✓</div><div>Waste Disposal Technician (Class A)</div><div>Certified 2026</div></div><div><div>○</div><div>Occupational Safety and Health Management Personnel (Level B)</div><div>2026 / 11 • mid-Nov exam scheduled</div></div><div><div>○</div><div>Occupational Safety Management Specialist (Level A)</div><div>2027 / 03 • exam scheduled</div></div></div><div>Instruments</div><div>Maintenance</div><div>TGA Q500</div><div>DSC 2500</div><div>Operation</div><div>ARES-G3</div><div>Anton Paar DMA 5002</div><div>Glovebox</div><div>Procurement</div><div>Glovebox O₂ sensor</div><div>DSC sample press</div><div>Hydraulic electrode punch</div><div>Vacuum pump</div><div>Eppendorf Pipette</div><div>Low-temperature bath</div></div>	
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